

Lubrication reliability analysis of wind turbine main bearing in random wind field

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ABSTRACT

This paper aims to establish a method to evaluate the lubrication reliability of the wind turbine main bearing in random wind fields. The aeroelastic simulations are conducted and the hub load is captured to generate the lubrication parameter database. Then, the lubrication reliability of the main bearing is evaluated and the effect of the random wind field on lubrication reliability is studied. Finally, an engineering formula to rapidly evaluate the lubrication failure probability for this wind turbine main bearing is obtained. Results show that the average wind speed and turbulence intensity affect the failure probability by S-shaped and linear law respectively. The lubrication of this wind turbine's main bearing almost completely fails when the average wind speed is 7 m/s.

1. Introduction

Renewable-based power generation technologies are established in consideration of climate change, environmental protection, and sustainable development. Wind energy is one of the fast-growing ones [1–3]. Wind turbines are exposed to extreme conditions and face the problem that components fail prematurely. As the wind turbines move further from shore, the length of the operating lifetime of wind turbines has an essential effect on the economics of wind power due to the high maintenance costs [4–6]. The annual failure frequency of the drivetrain and gearbox is lower than the electrical and electronic control systems [7]. However, due to the large size and heavy masses, the drivetrain and the gearbox are harder and more expensive to restore as they require cranes and other large equipment [5]. Therefore, the drivetrain and the gearbox are the leading cause of the downtime of the wind turbine.

Field experience shows that the main bearing used in the main shaft is highly prone to failure much earlier than the expected lifetime [7]. The early failure is because the main bearing supports the large mass of the rotor and the main shaft; hence it interacts with the rotor and drivetrain. This component undergoes the severe impact of harsh environmental conditions such as variable wind fields, air pressure, and random load. The failure of the main bearing may be costly as repairs require the entire turbine rotor to be removed and supported during changeovers. Therefore, the importance of bearing reliability is

increasingly prominent, leading to more focus on this component [8,9]. There are papers dealing specifically with the detection of the fault after it has occurred for the main bearing, which is modeled.

based on the vibration, bearing temperature, electric current, and other measured physical signals [10–12]. However, the mechanisms leading to premature main bearing failures cannot be explained. More research on the main bearing focuses on load modeling, load characteristics, and fatigue life prediction without considering lubrication conditions within it [13–15]. However, relative studies show that the wear which affects the main bearing service life is primarily caused by micropitting. And a poor lubrication condition leads to an insufficient local lubrication film thickness, allowing direct interaction of the asperities on rollers and raceway [16–18]. Therefore, lubrication significantly affects the main bearing's lifetime and must be considered as part of the analysis to study the working conditions, failure mechanisms, and reliability of the main bearing [7]. Some attempts have been conducted on the lubrication condition within the main bearing; unfortunately, the lubrication was not modeled directly [19,20].

The entire load pathway within the wind turbine contains the interactions between turbulent wind field and aerodynamic/control, drivetrain load balance, and load distribution within the main bearing. The incident wind field is time-varying, and the main bearing experiences large fluctuating load patterns throughout operation due to wind shear and turbulence [5]. Moreover, the high variability in the rotation speed of the rotor across a random wind field strongly affects the

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Nomenclature

I_{ref}	reference value of the turbulence intensity
K	total combined stiffness, N/m
$p_{failure}$	lubrication failure probability, %
Q_a	axial reaction force, N
Q_r	radial reaction force, N
Q_ψ	reaction force at angle ψ , N
U_{ave}	average wind speed, m/s
Z	number of rollers
δ	approach between inner and outer rings
δ_a	approach along the axial
δ_r	approach along the radial
σ	combined surface roughness
ψ	angle around the circumference
ψ_L	limiting angle

lubrication condition within the main bearing. It is foreseeable that the lubrication condition of the main bearing will fluctuate to a certain extent due to the wind field shear and turbulence. Whether the main bearing can always ensure well lubrication conditions is a question worthy of in-depth consideration and study. In other words, what is the lubrication reliability of the main bearing in a turbulent wind field? Answering this question requires a deeper understanding of the relationship between the random wind field and the lubrication, which is still unclear [21]. Therefore, there is a considerable knowledge gap regarding wind turbine main bearing lubrication, especially the lubrication reliability when considering the random wind field, indicating that further work is necessary to understand these relationships better.

In this paper, we aim to propose a method to evaluate the lubrication reliability of the wind turbine main bearing in random wind fields. The analysis followed the entire wind turbine load pathway from the incident wind field to the load distribution within the main bearing. On this basis, the effect of the average wind speed and wind turbulence intensity on the main bearing lubrication reliability is studied, establishing the relationship between the random wind field and the main bearing lubrication reliability. An engineering formula to rapidly evaluate the lubrication failure probability for this wind turbine main bearing is obtained.

2. Methodology

The lubrication reliability analysis for the main bearing under a random wind field includes four stage process. It starts from the aeroelastic simulations of the wind turbine, which generate the time series of hub load and shaft speed under random wind fields across a range of hub-height average wind speed and turbulence levels. Then the extracted time series load dataset is input to an analysis model to evaluate the applied load of the main bearing. The analysis is conducted at each timestep to ensure a static load balance. After that, a simplified double-row spherical roller bearing model is employed to analyze the internal load distribution within the main bearing such that the reaction forces of rollers can be found. Finally, the maximum roller load and main-shaft speed are input to the lubrication reliability analysis to obtain the lubrication state. The four-stage process will be described in more detail below.

2.1. Aeroelastic simulations of the wind turbine

TurbSim and OpenFast are open-source software, and their joint simulation has been widely used in wind turbine research [22]. TurbSim can generate stochastic, full-field, turbulent wind fields, it is employed here to generate the time-varying wind fields with different average

Table 1

Turbulence levels in IEC design standards.

Turbulence levels	Turbulence intensity I_{ref}
A	0.16
B	0.14
C	0.12

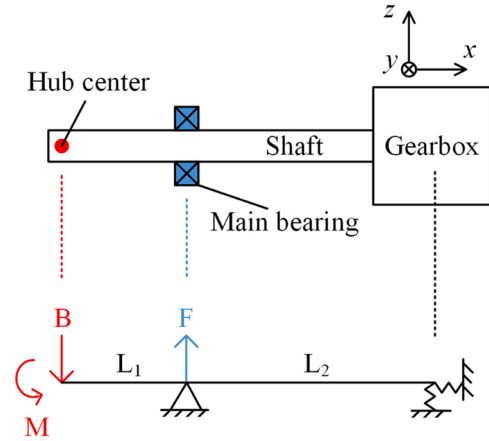


Fig. 1. Schematic and equivalent diagram of SMB configuration.

wind speeds and turbulence levels defined by the International Electrotechnical Commission (IEC) design standards (as shown in Table 1). And a normal turbulence model (NTM) is employed. The accuracy of OpenFast has been widely accepted and the aeroelastic simulations in this paper are simulated by using it [23]. The full-field wind files generated by TurbSim are inputted to OpenFast to conduct aeroelastic simulations. 160 Hz time series of hub load with six degrees of freedom was extracted (three forces and three moments) for each wind field. That is, channels RotThrust, LSShftFys, LSShftFzs, LSShftMxs, LSSTipMys and LSSTipMzs are outputted. These hub loads will be transmitted to the main bearing via the main shaft to evaluate the applied and internal distributed load for the main bearing in the following sections. The shaft speed is also extracted by the output channel RotSpeed to calculate the dimensionless lubrication parameters in the lubrication reliability analysis. The employed wind turbine is a publicly available 5-MW model, and summary information for it will be shown in Section 3. More detailed design parameters can be found in the definition document [24].

2.2. Evaluation of the main bearing applied load

The current work needs to determine bearing loads across a large number of timesteps to analyze the lubrication reliability. An overly complex model will bring an unacceptable amount of computation, so the load applied on the main bearing is evaluated by a statically determinate main bearing model [25]. A 160 Hz time series of a 10-min long wind field results in 96,000 data points. To avoid unwieldy data quantities, 2000 timesteps are extracted from each 10-min time series to form the dataset. Preliminary analysis shows that this data amount is sufficient to characterize the wind field.

There are usually two hub and main bearing configurations. The first configuration features a single main bearing (SMB), and the second one features double main bearing rows spaced mounting on either side of the hub or both downwind of the hub (DMB). This paper focuses on the SMB (as shown in Fig. 1) case but is not limited to the hub and main bearing configurations. The simple theory can describe it and each in-plane force-moment pair is statically balanced at each timestep. The three-dimensional hub load series in six degrees of freedom is decomposed into orthogonal components. For a given axis, the balancing moments of

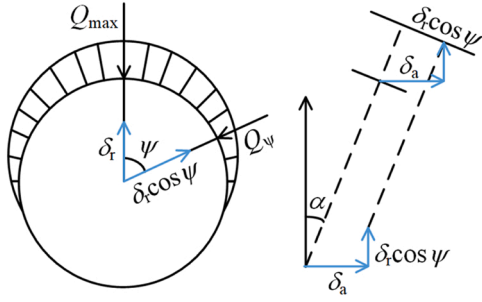


Fig. 2. Displacement geometries in a spherical roller-bearing row.

the gearbox can be statically determined and expressed as Eq. (1), and then the main bearing reaction force can be obtained by Eq. (2). After determining the reaction forces along the y and z axis respectively, the total radial reaction force of the main bearing can be synthesized as Eq. (3), while the axial reaction force is opposite to the rotor thrust. It should be noted that the required load in the bearing internal load distribution analysis in Section 2.3 is that being applied to the bearing by the shaft, which has an opposite direction to the reaction force.

$$M_i + (L_1 + L_2)B_i - L_2F_i = 0 \quad (1)$$

$$F_i = \frac{M_i + (L_1 + L_2)B_i}{L_2} \quad (2)$$

$$F_r = \sqrt{(F_y)^2 + (F_z)^2} \quad (3)$$

where i denotes a given axis y or z , M_i denotes the moments about the i -axis, B_i denotes the input forces and F_i is bearing reaction forces along the i -axis, L_1 and L_2 denotes the length from the main bearing to the hub center and the gearbox center respectively, F_y and F_z are bearing reaction forces along y -axis and z -axis respectively.

2.3. Internal distributed load for the main bearing

A double-row spherical roller bearing is equipped in the analyzed wind turbine. According to Hart et al. [26], under combined radially and axially applied load, the displacement geometries in a spherical roller bearing row can be presented in Fig. 2. Around the circumference, the approach between the inner and outer rings δ_ψ at an angle ψ can be expressed as Eq. (4). The spherical rollers can be treated as compression-only non-linear springs acting between raceways. The relationship between the reaction force and the undergoing deformation can be expressed as Eq. (5). The calculation of the total combined stiffness can be found in Hart et al. [26]. According to Eq. (4) and Eq. (5), the reaction force around circumference can be derived as Eq. (6). In fact, not all the rollers are underloaded for specific deformation, and the limiting angle ψ_L reflecting the loaded area is $\psi_L = \arccos(-\delta_a \tan \alpha / \delta_r)$. The reaction force at a point on the bearing circumference can be decomposed into the radial and axial components. The total reaction forces in the two directions, radial reaction force Q_r and axial reaction force Q_a , can be calculated by integrating Q_ψ at each point on the circumference as Eq. (7). The bearing reaction force and the applied load are balanced under the equilibrium assumption. The displacement problem can be solved by finding the solution of Eq. (8). Under the double-row case, the relationships in Eq. (9) should also be considered. After the above problem is solved, the values of δ_r and δ_a can be found, and the load distribution within the main bearing around the circumference can be determined as Eq. (10).

$$\delta_\psi = \delta_{\max} \left[1 - \frac{1}{2\varepsilon} (1 - \cos \psi) \right] \quad (4)$$

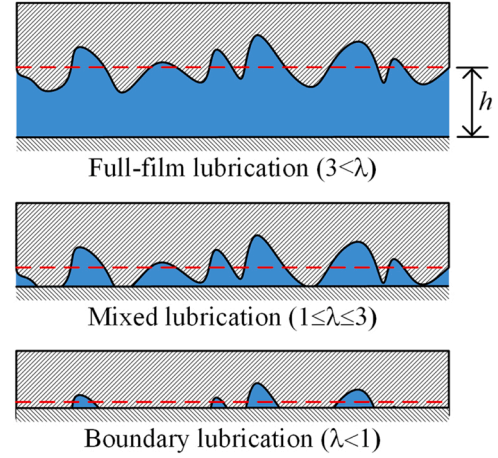


Fig. 3. Lubrication regimes with different film thickness ratio.

$$Q = K\delta^{3/2} \quad (5)$$

$$Q_\psi = Q_{\max} \left[1 - \frac{1}{2\varepsilon} (1 - \cos \psi) \right]^{3/2} \quad (6)$$

$$Q_r = \frac{ZQ_{\max} \cos \alpha}{2\pi} \int_{-\psi_L}^{\psi_L} \left[1 - \frac{1}{2\varepsilon} (1 - \cos \psi) \right]^{3/2} \cos \psi d\psi \quad (7)$$

$$Q_a = \frac{ZQ_{\max} \sin \alpha}{2\pi} \int_{-\psi_L}^{\psi_L} \left[1 - \frac{1}{2\varepsilon} (1 - \cos \psi) \right]^{3/2} d\psi$$

$$(F_r - Q_r)^2 + (F_a - Q_a)^2 = 0 \quad (8)$$

$$\delta_{r1} = \delta_{r2} = \delta_r, -\delta_{a1} = \delta_{a2} = \delta_a, Q_r = Q_{r1} + Q_{r2}, Q_a = Q_{a2} - Q_{a1} \quad (9)$$

$$Q_r = \begin{cases} \frac{ZQ_{\max}}{2\pi} \left[1 - \frac{1}{2\varepsilon} (1 - \cos \psi) \right]^{3/2} & -\psi_L < \psi < \psi_L \\ 0 & \text{otherwise} \end{cases} \quad (10)$$

where $\delta_{\max} = \delta_a \sin \alpha + \delta_r \cos \alpha$, δ_r and δ_a are approach along the radial and axial respectively, α is the contact angle, ψ is the angle around the circumference, and $\varepsilon = 0.5[1 + (\delta_a \tan \alpha) / \delta_r]$, K is the total combined stiffness of roller-raceway interfaces, Z is the number of rollers in a bearing row, 1 and 2 denote the upwind and downwind side row respectively.

2.4. Lubrication reliability model

The lubrication state of the main bearing can be evaluated by the lubrication reliability method [27]. Lubrication can be divided into different regimes according to the film thickness ratio λ , as shown in Fig. 3. A full-film lubrication can completely separate the contact surfaces and avoid direct interaction between asperities. Full-film lubrication is beneficial for reducing wear, friction, and the probability of failures such as pitting and gluing and is, therefore, named reliable lubrication [27]. However, when the full-film lubrication fails into mixed or boundary lubrication (generally considered as film thickness ratio $\lambda < 3$), direct contact of the asperities occurs with the potential for surface damage. Pei et al. defined the failure of the full-film lubrication as the lubrication failure [27]. The power function of the full-film lubrication can be expressed as Eq. (11), where $h_{\min}$ is the minimum film thickness. Wu et al. proposed the film thickness prediction formula of grease lubrication considering the roughness effect [28]. Here, this W-H equation is employed to predict the dimensionless grease film thickness, as shown in Eq. (12). Previous studies have shown that the dynamic EHL effects can be negligible in the direction of rolling for wind

Table 2
Parameters for the 5-MW wind turbine.

Parameters	Values
Hub height	90 m
Mass of rotor	110,000 kg
Mass of blade	53,220 kg
Mass of hub	56,780 kg
Distance from hub center to main bearing(L_1)	0.8 m
Distance from gearbox to main bearing(L_2)	2.5 m

Table 3
Main parameters of the turbulent wind fields.

Wind field	Average wind speed U_{ave}	Turbulence intensity I_{ref}
8-B	8 m/s	0.14(B-level)
10-B	10 m/s	0.14(B-level)
12-B	12 m/s	0.14(B-level)
10-C	10 m/s	0.12(C-level)
10-A	10 m/s	0.16(A-level)

Table 4
Parameters of the main bearing and lubricant.

Parameters	Values
Combined surface roughness of roller-raceway interfaces	≈ 400 nm
Equivalent contact radius	$R_{in} = 0.0399$ m, $R_{out} = 0.0481$ m
Equivalent elastic modulus	225.3 GPa
Plastic viscosity	5.3 Pa·s ⁿ
Rheological index	0.87

turbine main bearing [25]. In addition, Cen et al. pointed out that the film thickness was not significantly affected by introducing a typical SRR (slide-to-roll ratio) for rolling bearings [29]. Hence, although the operating condition of the M-K equation is considered to be steady-state and pure rolling, it is still applicable for EHL analysis of the wind turbine main bearing in random wind fields.

$$g_r(h_{min}, \sigma) = h_{min} - 3\sigma \quad (11)$$

$$H_{min} = 11.954n^{20.714}W^{-0.126}U^{0.693n^{1.638}}G^{0.345n^{7.526}} \times (1 + 0.0009n^{0.274}\bar{\sigma}^{3.449n^{6.171}}W^{0.192}U^{-1.576n^{5.050}}G^{0.514n^{-1.471}}) \quad (12)$$

where σ is the combined surface roughness, n is the rheological index of grease, H_{min} , W , U , V , G and $\bar{\sigma}$ are dimensionless parameters as defined in

Wu et al. [28].

For a time-varying wind field, after obtaining the minimum film thickness at each timestep, the failure probability of lubrication $p_{failure}$ can be calculated as:

$$p_{failure} = \Pr[g_r(h_{min}, \sigma) \leq 0] = \Pr(\lambda \leq 3) \quad (13)$$

3. Example and analysis

In this section, a publicly-available 5-MW wind turbine model is taken as an example for analysis [24], and the summary information for it is shown in Table 2. The control strategy of the wind turbine determines the rotor speed and thrust force at a given wind speed. This influences the main bearing's rolling speed and the load condition, thus affecting its lubrication state. This 5-MW wind turbine employs the conventional variable-speed, variable blade-pitch-to-feather configuration. The control strategy is implemented based on two control systems: a generator-torque controller and a full-span rotor-collective blade-pitch controller. The functions of the generator-torque controller and the blade-pitch controller are to maximize power capture below the rated operation point and regulate generator speed above the rated operation point, respectively. More detailed design parameters can be found in the definition document [24]. 3D turbulent wind fields were generated in compliance with IEC standards using a Kaimal spectrum. The main parameters of typical wind fields are shown in Table 3, where A, B, and C represent the turbulence intensity defined by IEC (as shown in Table 1).

The wind turbine was simulated in each turbulent wind field, and the hub load and the shaft speed were extracted to analyze the main bearing load. The analyzed wind turbine employs an SMB configuration, and the main bearing is a double-row spherical roller type, an SKF 230/750CA/W33. Some of its geometric information is available on the SKF official website and is shown in Table 4; however, more detailed parameters are proprietary and not presented. The rheological parameters of grease are obtained through rheological experiments and fitted into the Ostwald fluid model, the parameters are shown in Table 4. The analysis follows the process described in Section 2.

4. Results and discussion

In different installation areas, the wind fields will also vary greatly, which may change the lubrication performance of the wind turbine main bearing. In this section, the result of lubrication reliability is discussed, and the effect of the average wind speed and turbulence intensity on main bearing lubrication reliability is studied.

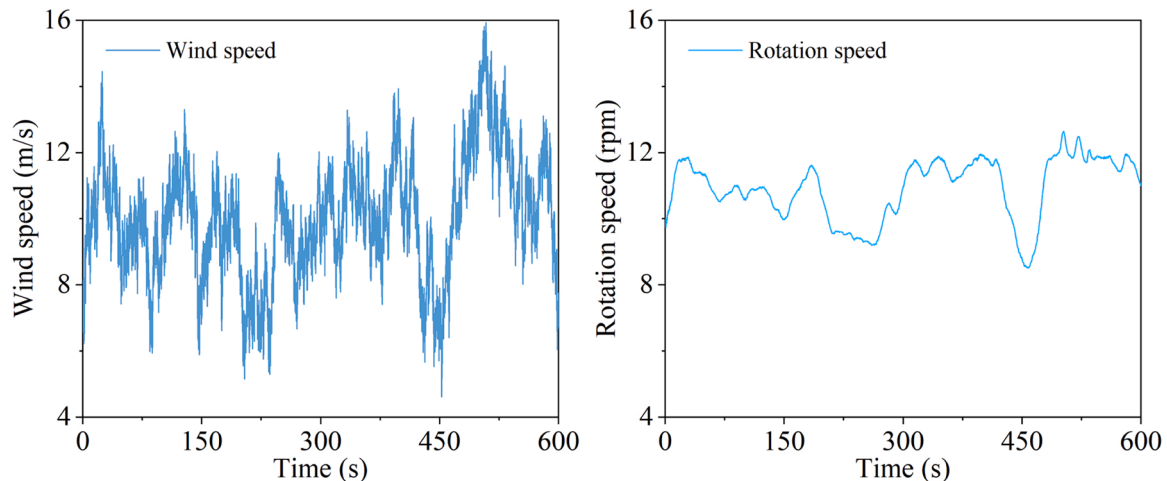


Fig. 4. Rotation speed of the rotor and main bearing in 10-B wind field (a) time-varying wind speed in x direction (b) the rotation speed.

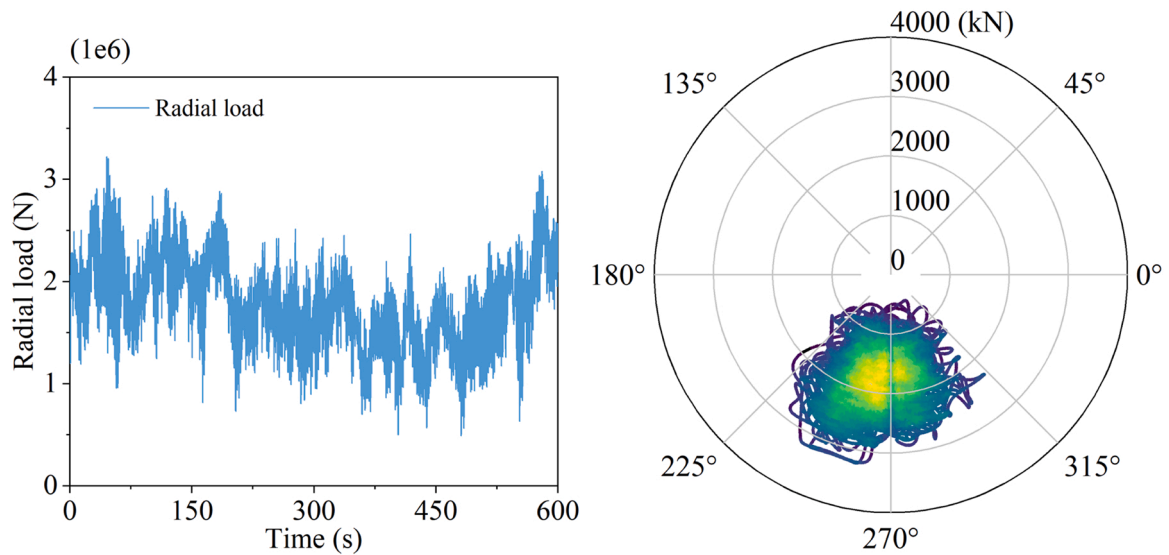


Fig. 5. Radial load from 10-min aeroelastic simulations in 10-B turbulent wind field (a) time-varying applied radial load (b) density plots of applied radial load.

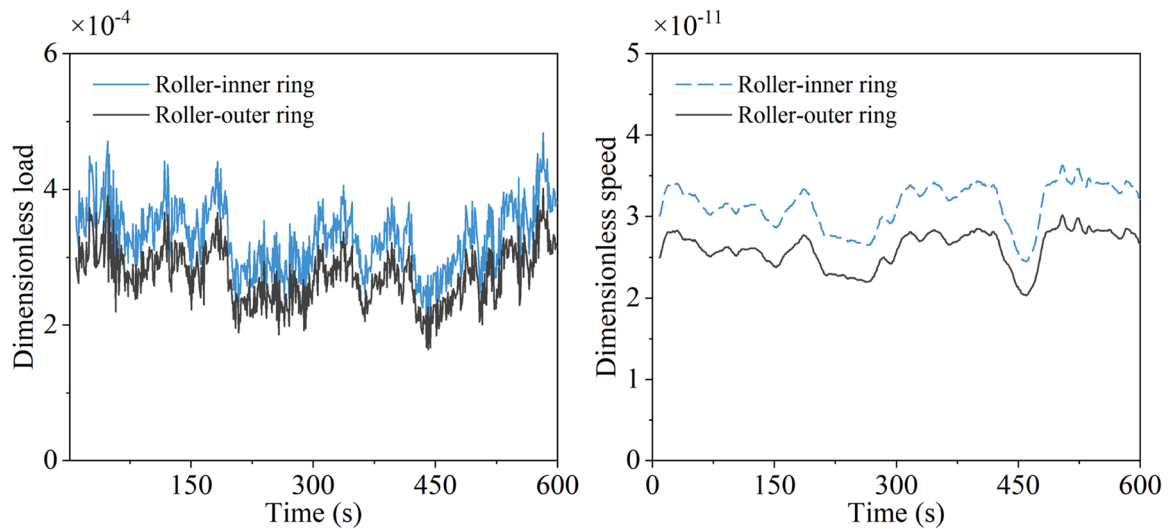


Fig. 6. Dimensionless lubrication parameters (a) dimensionless load (b) dimensionless speed.

4.1. Lubrication reliability of the main bearing

In this section, the result of the lubrication reliability analysis of the main bearing in the 10-B wind field (10 m/s average wind speed and middle turbulence intensity) is illustrated. Following the process described in Section 2, the aeroelastic simulation is conducted first, the wind speed and the rotor speed are shown in Fig. 4. It can be observed that the rotor speed fluctuations are smaller than wind speed fluctuations, which is contributed to the control system keeps the rotor speed changing smoothly in the random wind field. The applied load on the main bearing can be obtained after the statically determinate analysis and the internal load analysis, and the results are shown in Fig. 5. The applied loads shown in the figures are those looking towards the turbine from the upwind side. Fig. 5(a) shows that the radial load obviously fluctuation in the random wind field. Fig. 5(b) is the density plot whose points are colored according to the probability of occurrence of the load. It is drawn in polar coordinates and can reflect the direction distribution of the applied load around the bearing circumstance. Fig. 5(b) shows that with visible fluctuations, the radial load centered around the rotor weight in the vertical direction.

The applied load is input into the analysis model in Section 2.3 to

calculate the internal distributed load for the main bearing. During the revolution around the bearing circumference, the rollers experience a continuous change in load level from minimum to maximum but have the same rolling speed. Previous studies show a higher load is more unfavorable for lubrication [30,31]. Therefore, for the whole bearing, the position under the maximum load has the worst lubrication state (with the same speed, material, and roughness). Regardless of how the load is distributed along the bearing circumference, the roller with the maximum load is always the focus. The maximum load is used as the input parameter to the lubrication model, while the rolling speed is the same for different rollers around the whole bearing circumference. According to Section 2.4, dimensionless parameters are calculated. Fig. 6 shows that the dimensionless load fluctuates more strongly than the dimensionless speed. This is because the wind turbine's control system keeps the rotor speed changing smoothly in the turbulent wind field. It can be observed that the roller-inner ring contact has an enormous dimensionless load and dimensionless speed. This can be explained by the difference in the equivalent radii of the roller-inner ring and roller-outer ring contact.

All the dimensionless lubrication parameters are entered into the lubrication reliability model to calculate the minimum film thickness

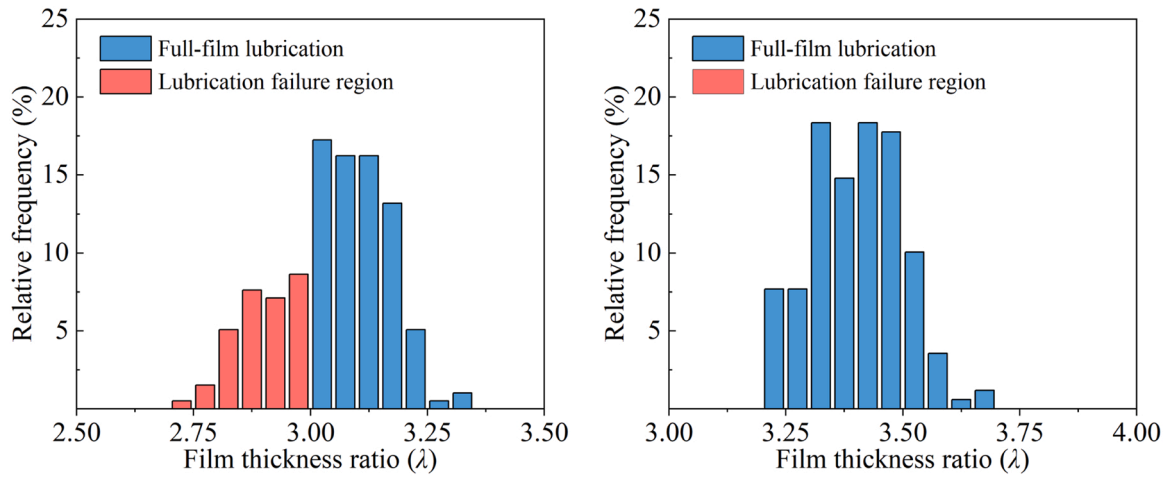


Fig. 7. Probability distribution of the film thickness ratio (a) roller-inner ring contact (b) roller-outer ring contact.

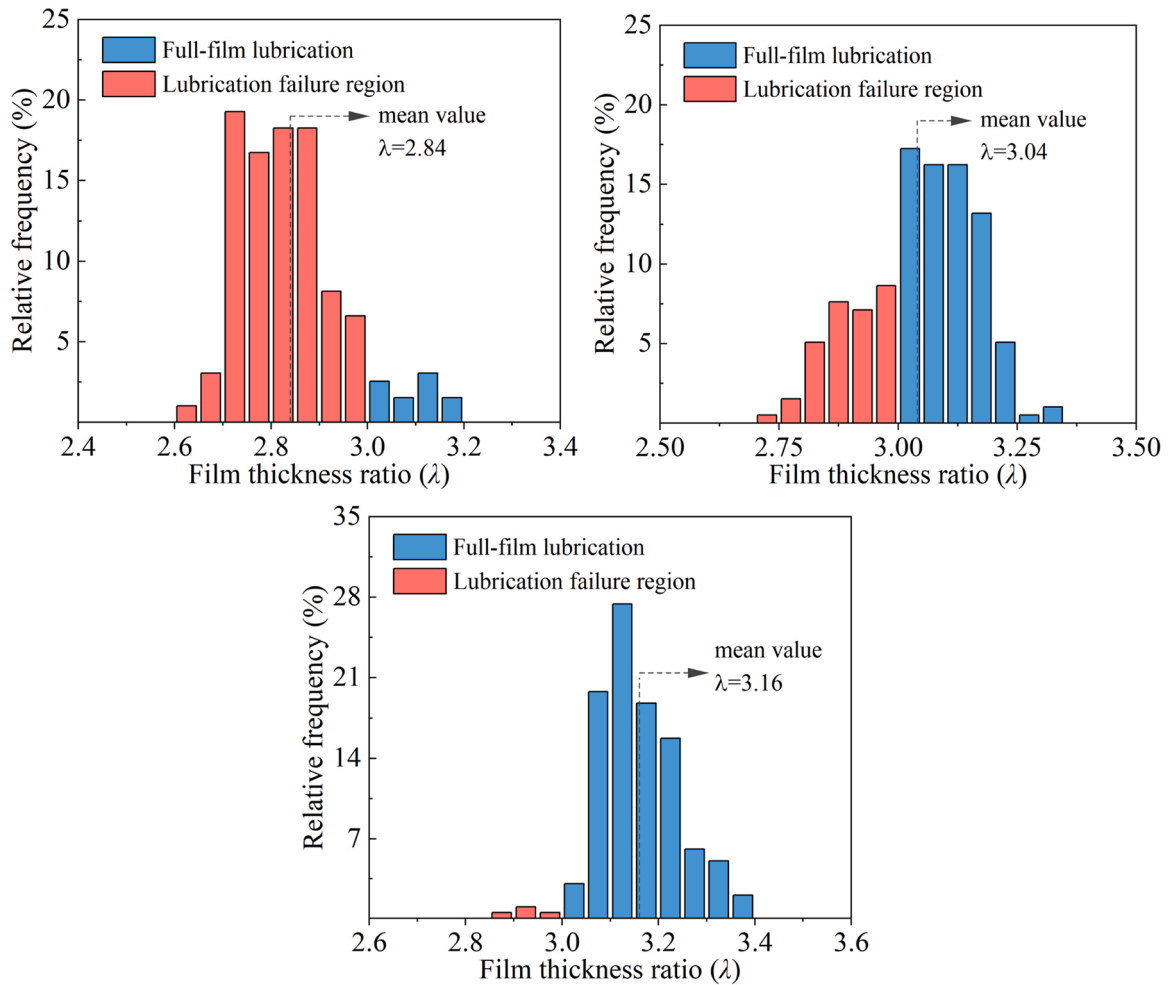


Fig. 8. Probability distribution of the film thickness ratio in wind fields with different average wind speeds (a) 8-B wind field with 8 m/s average wind speed (b) 10-B wind field with 10 m/s average wind speed (c) 12-B wind field with 12 m/s average wind speed.

and film thickness ratio. Statistical analysis of the film thickness ratio is carried out, and the probability distribution of it is shown in Fig. 7. According to Eq. (13), the lubrication failure is essentially the failure of the full-film thickness lubrication, which commonly occurs when the film thickness ratio $\lambda \leq 3$ [32–34]. Therefore, the failure probability of lubrication is equal to the relative frequency of the lubrication failure

region in Fig. 7. After statistical analysis, the lubrication failure probability of the main bearing roller-inner ring and roller-outer ring contact are 30.46% and 0, respectively. The result indicates that the roller-inner ring contact is more likely to be in a mixed lubrication state and, thus, more prone to damage. The above analysis shows that for this bearing, the lubrication condition of the inner raceway is worse and has a higher

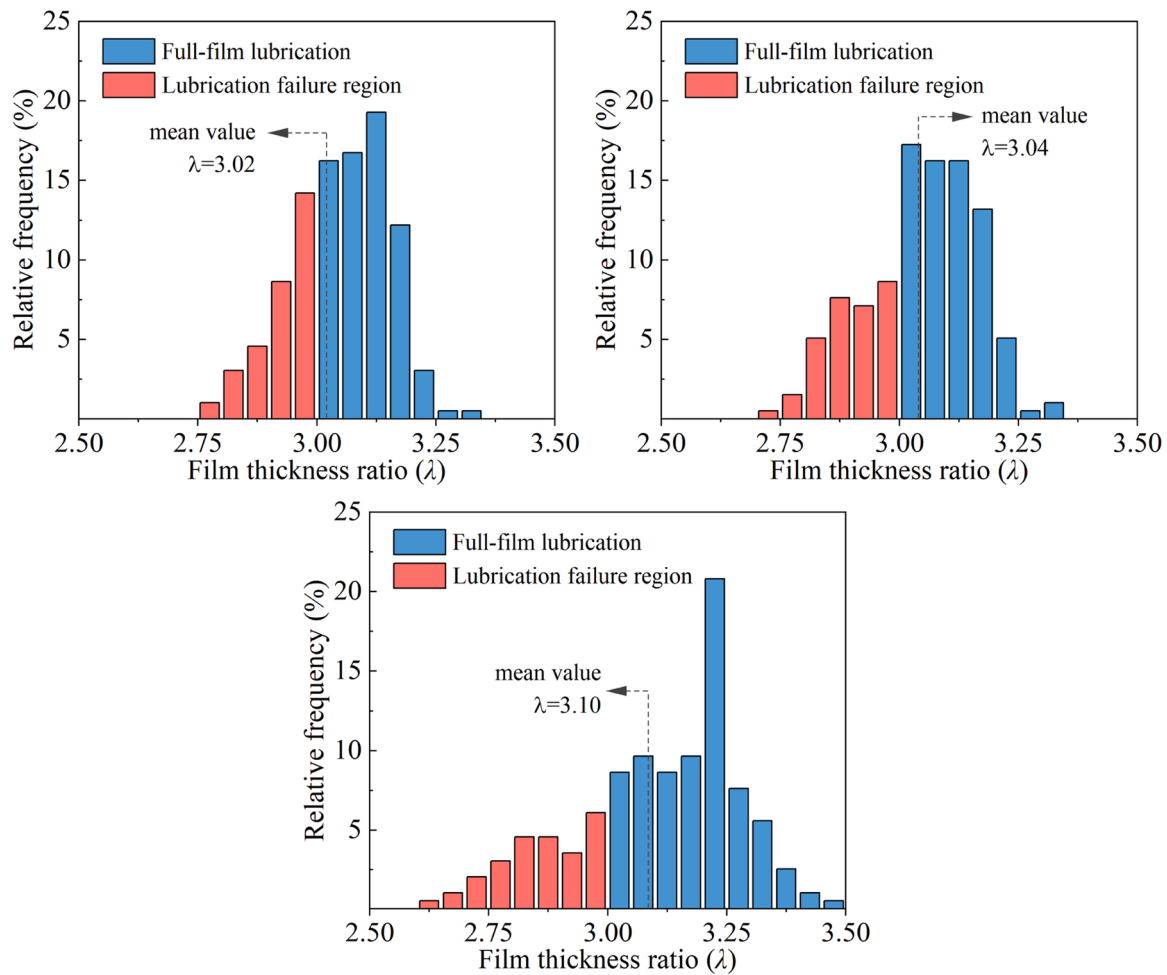


Fig. 9. Probability distribution of the film thickness ratio in wind fields with different turbulence intensity (a) 10-C wind field with 0.12 turbulence intensity (b) 10-B wind field with 0.14 turbulence intensity (c) 10-A wind field with 0.16 turbulence intensity.

failure probability. Therefore, the subsequent analysis mainly focuses on the inner raceway.

4.2. Effect of average wind speed on lubrication reliability

In this section, the effect of average wind speed on lubrication reliability is studied. Three wind fields (8-B, 10-B, and 12-B wind fields in Table 3) with different average wind speed levels (8 m/s, 10 m/s, and 12 m/s) but with identical turbulence intensity (B-level) are employed to simulate the aeroelastic of the wind turbine. The load analysis and the lubrication reliability analysis of the main bearing are then conducted according to the process in Section 2, and the results are shown in Fig. 8.

It can be observed from Fig. 8 that for different average wind speeds, the probability distribution of film thickness ratio shows significant differences. In the wind field with a larger average wind speed, the mean value of the film thickness ratio is larger and the overall level of the bearing film thickness ratio is higher. This means that the 10-min overall lubrication performance of the main bearing is better in the random wind field with a higher average wind speed. Further statistical analysis showed that the probability of lubrication failure in Fig. 8(a)–(c) is 91.37 %, 30.46 %, and 2.03 %, respectively, showing that the average wind speed obviously affects the lubrication reliability and the failure probability. This is due to a higher wind speed means a larger dimensionless speed, and the film thickness quite sensitively increases with it [35,36]. Although different average speeds of the wind field also affect the bearing load distribution, the sensitivity of the lubrication state to the load is much less than that to the speed. Therefore, the probability of

lubrication failure is closely related to the average wind speed and decreases with its increase.

The above analysis shows that the lubrication performance of the wind turbine main bearing is significantly affected by the average wind speed. To ensure the service life of the wind turbine, the structure of the main bearing and the wind turbine should match the characteristics of the wind field where it is installed.

4.3. Effect of wind turbulence intensity on lubrication reliability

In this section, the effect of wind turbulence intensity on lubrication reliability is analyzed. In wind fields with the same average wind speed (10 m/s) but different turbulence intensities (10-A, 10-B, and 10-C in Table 3), the lubrication performance of wind turbines was investigated. The probability distribution of the film thickness ratio is shown in Fig. 9. Results show that in wind fields with different turbulence intensities, the mean level of the main bearing's film thickness ratio is very similar, around 3. This means that from a long-term perspective, the turbulence intensity has little effect on the mean level of lubrication performance. This is because the three wind fields have identical average wind speeds. As mentioned before, the average wind speed is decisive for the overall level of lubrication performance.

However, the statistical analysis shows that the failure probability of the lubrication in the three wind fields is different, which are 29.47 %, 30.46 %, and 34.52 %, respectively. The failure probability increases as the wind turbulence intensity. Fig. 9 shows that the distribution range of the film thickness ratio becomes wider. This is because a larger

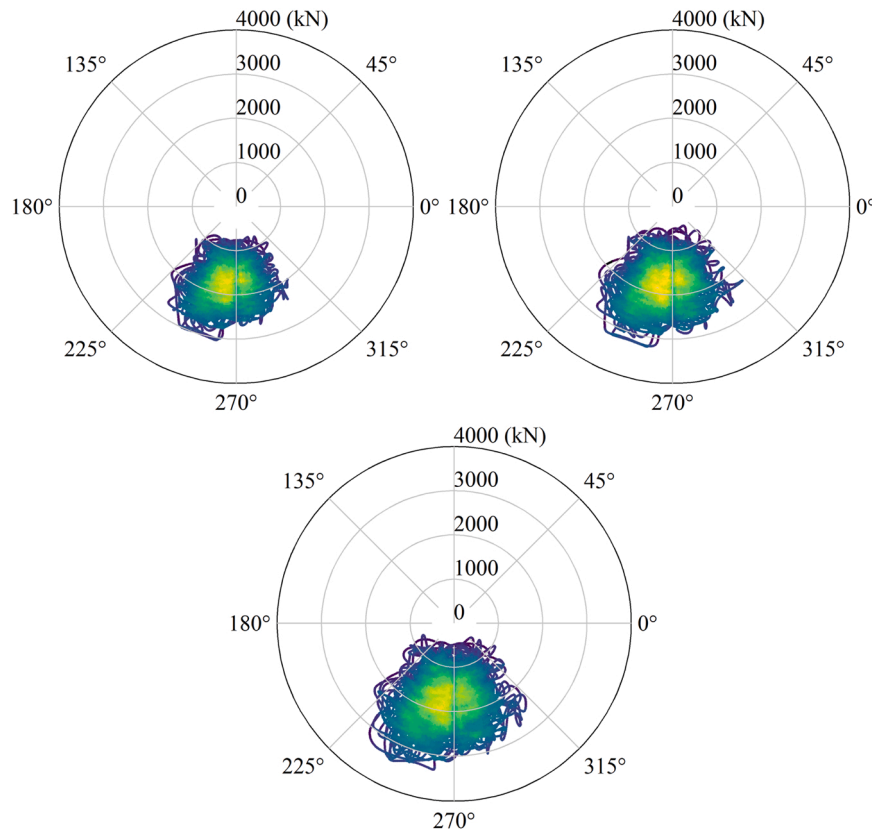


Fig. 10. Radial load for the main bearing in wind field with different turbulence intensity (a) 10-C wind field with 0.12 turbulence intensity (b) 10-B wind field with 0.14 turbulence intensity (c) 10-A wind field with 0.16 turbulence intensity.

Table 5
Simulation and curve fitting results of lubrication failure reliability.

Input		Failure probability (%)		
Average wind speed U_{ave} (m/s)	Turbulence intensity I_{ref}	Simulation	Curve fit	Error %
7	0.14	99.15	100.00	0.85
8		94.75	93.11	1.64
9		78.17	76.28	1.89
10		43.82	48.27	4.45
11		17.52	16.89	0.63
12		4.57	1.52	3.05
13	0.10	2.37	0.01	2.36
		39.76	45.50	5.74
		42.47	46.89	4.42
10		43.82	48.27	4.45
		44.50	49.66	5.16
	0.18	45.31	51.05	5.74

turbulence intensity enhances the dispersion of the wind speed. In this case, more low wind speeds and, thus, more little dimensionless speed parameters occur. The main bearing has worse lubrication performance with these low-speed parameters and the lubrication failure probability is higher. In addition to the effect of the low rolling speed, the degree of applied load fluctuation is also increased due to a larger turbulence intensity, as shown in Fig. 10. This leads to more extremely small loads, which can deteriorate the lubrication performance. The combined effect of the above factors enhances the dispersion of bearing lubricating film thickness, occurring more lubrication failure points.

4.4. Evaluation formula for the lubrication failure probability

Predicting the lubrication failure probability is of great significance

for the evaluation of the wind turbine structure and wind field in engineering. In this section, an engineering formula to rapidly evaluate the lubrication failure probability for this wind turbine main bearing is obtained. To derive an appropriate curve-fitting formula, a variety of wind fields with different average wind speeds and turbulence intensities are generated. For each wind field with identical average wind speed and turbulence intensity, 6 different random wind fields are generated. In these wind fields, the lubrication failure probability of the wind turbine main bearing is evaluated, and the average results of 6 random wind fields are tabulated in Table 5. The law of lubrication failure probability changing with the characteristics of the wind field is shown in Fig. 11, which is plotted using the average results of 6 random wind fields with the error bars. Results show that when the average speed of the wind field is equal to 7 m/s, the probability of lubrication failure is close to 100 %. While the average wind speed reaches 13 m/s, the lubrication failure almost does not occur, which means that the main bearing is almost always in the full-film lubrication state. The lubrication failure probability changes with average wind speed U_{ave} in the form of an S-shaped curve, as shown in Fig. 11 (a). The Gompertz model is widely used to fit the S-shaped curve and can be employed here to describe the effect of average wind speed on the lubrication failure probability in the form of $a \times b^{cU_{ave}}$ [37]. In addition, the lubrication failure probability changes approximately linearly with the turbulence intensity I_{ref} , as shown in Fig. 11 (b). Therefore, the effect of the turbulence intensity I_{ref} on the lubrication failure probability can be fitted in the form of $d \times (I_{ref} - e)$. Considering both the above relationships, the final form of the curve-fitting equation is assumed as $p_{failure} = a \times b^{cU_{ave}} + d \times (I_{ref} - e)$, where a, b, c, d, e are unknown constants to be determined.

After fitting the data in Table 5, the prediction formula of lubrication failure probability for this wind turbine main bearing is obtained as Eq. (14).

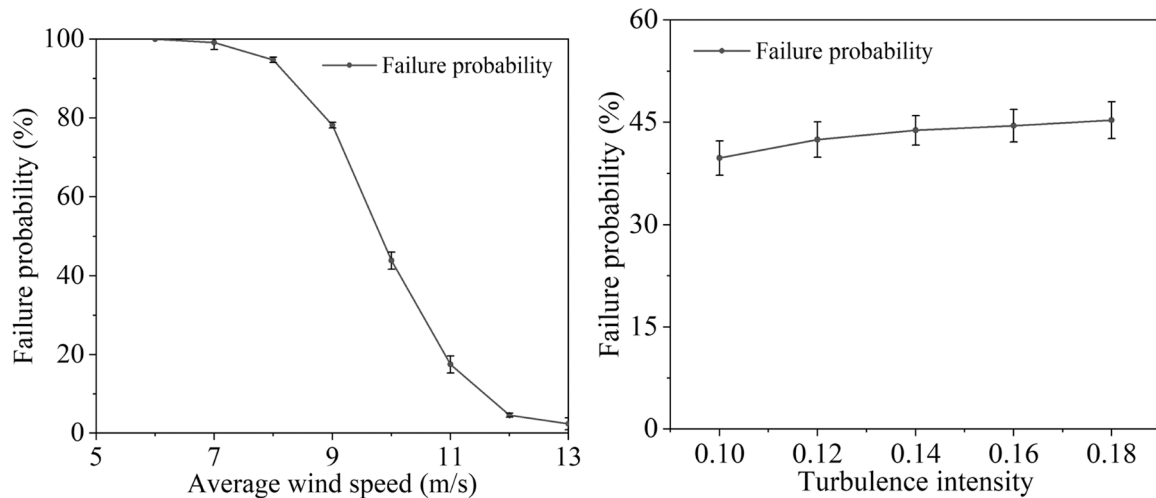


Fig. 11. Lubrication failure probability in different wind fields (a) lubrication failure probability with average wind speed varies (b) lubrication failure probability with turbulence intensity varies.

$$p_{\text{failure}} = 108.6 \times \left(0.9998^{2.295U_{\text{ave}}}\right) + 69.291 \times (I_{\text{ref}} - 0.14) \quad (14)$$

where U_{ave} is the average wind speed, and I_{ref} is the turbulence intensity.

Errors between simulation results and the fitted curve are listed in Table 5. The error analysis shows that the fitted curve has the ideal accuracy. It should be noted that the value range of p_{failure} is between 0 and 1. When the value of Eq. (14) is greater than 100 %, it should be taken as 100 %, and when it is less than 0, it should be taken as 0.

Eq. (14) can evaluate the lubrication failure probability of wind turbine main bearing in random wind fields. The wind field suitable for this type of wind turbine shall be selected according to this lubrication failure prediction formula. We emphasize here that this prediction formula applies to the wind turbine structure and lubricant type discussed in this paper, but the established method is universal and can be used to study other types of wind turbines, main bearings and lubricants. Existing study shows that there is significant thermal inertia in the main bearing [38]. This means the temperature of the main bearing only has a relatively weak correlation with the load and rolling speed and is difficult to be predicted in random wind fields. Therefore, the thermal effect is not included here. It is emphasized that if starvation exists in reality, the film thickness and hence the lubrication failure probability should be modified according to the starvation degree.

5. Conclusion

In this paper, a method to evaluate the lubrication reliability of the wind turbine main bearing in random wind fields is proposed. The effect of wind field characteristics (average wind speed, turbulence intensity) on the lubrication reliability of a wind turbine main bearing is investigated. In addition, an engineering formula to rapidly evaluate the lubrication failure probability for this wind turbine main bearing is obtained. The following conclusions can be drawn:

- 1) For this wind turbine, the lubrication failure probability of the main bearing roller-inner ring contact is larger than that of the roller-outer ring contact. Therefore, the inner raceway is more likely to be in a mixed or boundary lubrication state and is more prone to damage.
- 2) The average wind speed obviously affects the overall level of lubrication reliability. The main bearing lubrication failure probability increases with the decrease in the average wind speed, showing an S-shaped curve. Larger turbulence intensity will enhance the dispersion of rolling speed and the dispersion of load, increase the probability of extreme working conditions, and thus affect the lubrication

reliability linearly. However, the turbulence intensity has little effect on the overall level of the lubrication performance of the main bearing from a long-term perspective.

- 3) When the average wind speed is about 7 m/s, the main bearing lubrication almost completely fails. In this case, this wind turbine main bearing works under a mixed even boundary lubrication regime, which may cause wear and other damages. When the average wind speed reaches 13 m/s, the lubrication is almost 100 % reliable.
- 4) An engineering formula to rapidly evaluate the lubrication failure probability for this wind turbine main bearing is obtained, which can be employed to evaluate the rationality of the wind turbine structure and installation position. If starvation exists, the film thickness and hence the lubrication failure probability should be modified according to the starvation degree.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data Availability

Data will be made available on request.

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